keep it simple science
ABN 54 406 994 557

PO Box 2575
PORT MACQUARIE NSW 2444
(02) 6583 4333 FAX (02) 6583 9467
www.keepitsimplescience.com.au
mail@keepitsimplescience.com.au

keep it simple science
Photocopy Master Sheets

Years 7-8

Solids, Liquids & Gases

Disk filename = "03.Matter"

Usage & copying is permitted according to the
Site Licence Conditions
only

Site Licence Conditions

A school (or other recognised educational institution) may store the disk contents in multiple computers (or other data retrieval systems) to facilitate the following usages of the disk contents:

• School staff may print unlimited copies on paper and/or make unlimited photocopies at one school and campus only, for use by students enrolled at that school and campus only, for non-profit, educational use only.

• School staff may use the disk contents to make audio-visual displays, such as via computer networks, or by using data projectors or overhead projectors, at one school and campus only, for viewing by students enrolled at that school and campus only, for non-profit, educational use only.

• School staff may allow students enrolled at that school and campus only to obtain copies of the disk files and store them in each student's personal computer for non-profit, educational use only.

IN SUCH CASE, THE SCHOOL SHOULD MAKE PARTICIPATING STUDENTS AWARE OF THESE SITE LICENCE CONDITIONS AND ADVISE THEM THAT COPYING OF DATA FILES BY STUDENTS MAY CONSTITUTE AN ILLEGAL ACT.

• In every usage of the disk files, the KISS logo and copyright declaration must be included on each page, slide or frame.

Please Respect Our Rights Under Copyright Law

Topics Available

Year 7-8 General Science

Disk Filename	Topic Name
01.Energy	Energy
02.Forces	Forces
03.Matter	Solids, Liquids & Gases
04.Mixtures	Separating Mixtures
05.Elements	Elements & Compounds
06.Cells	Living Cells
07.Life	Living Things
08.LifeSystems	Plant & Animal Systems
09.Astronomy	Astronomy
10.Earth	The Earth
11.Ecosystems	Ecosystems

Year 9-10 General Science

Disk Filename	Topic Name
12.Waves	Wave Energy (inc. Light)
13.Motion	Forces & Motion
14.Electricity	Electricity
15.Atoms	Atoms & Elements
16.Reactions	Compounds & Reactions
17.DNA	Cell Division & DNA
18.Evolution	Evolution of Life
19.Health	Health & Reproduction
20.Universe	The Universe
21.EarthScience	Earth Science
22.Resources	Resources & Technology

Year 11-12 Science Courses

Biology

Preliminary Core
Local Ecosystem
Patterns in Nature
Life on Earth
Evolution Aust. Biota
HSC Core
Maintain. a Balance
Blueprint of Life
Search for Better Health
Options
Communication
Genetics:Code Broken?

Chemistry

Preliminary Core
Chemical Earth
Metals
Water
Energy
HSC Core
Production of Materials
Acidic Environment
Chem.Monit.&Mngment
Options
Shipwrecks, Corrosion...
Industrial Chemistry

Earth & Envir. Science

Preliminary Core
Planet Earth...
Local Environment
Water Issues
Dynamic Earth
HSC Core
Tectonic Impacts
Environs thru Time
Caring for the Country
Option
Introduced Species

Physics

Preliminary Core
World Communicates
Electrical Energy...
Moving About
Cosmic Engine
HSC Core
Space
Motors & Generators
Ideas to Implementation
Options
Quanta to Quarks
Astrophysics

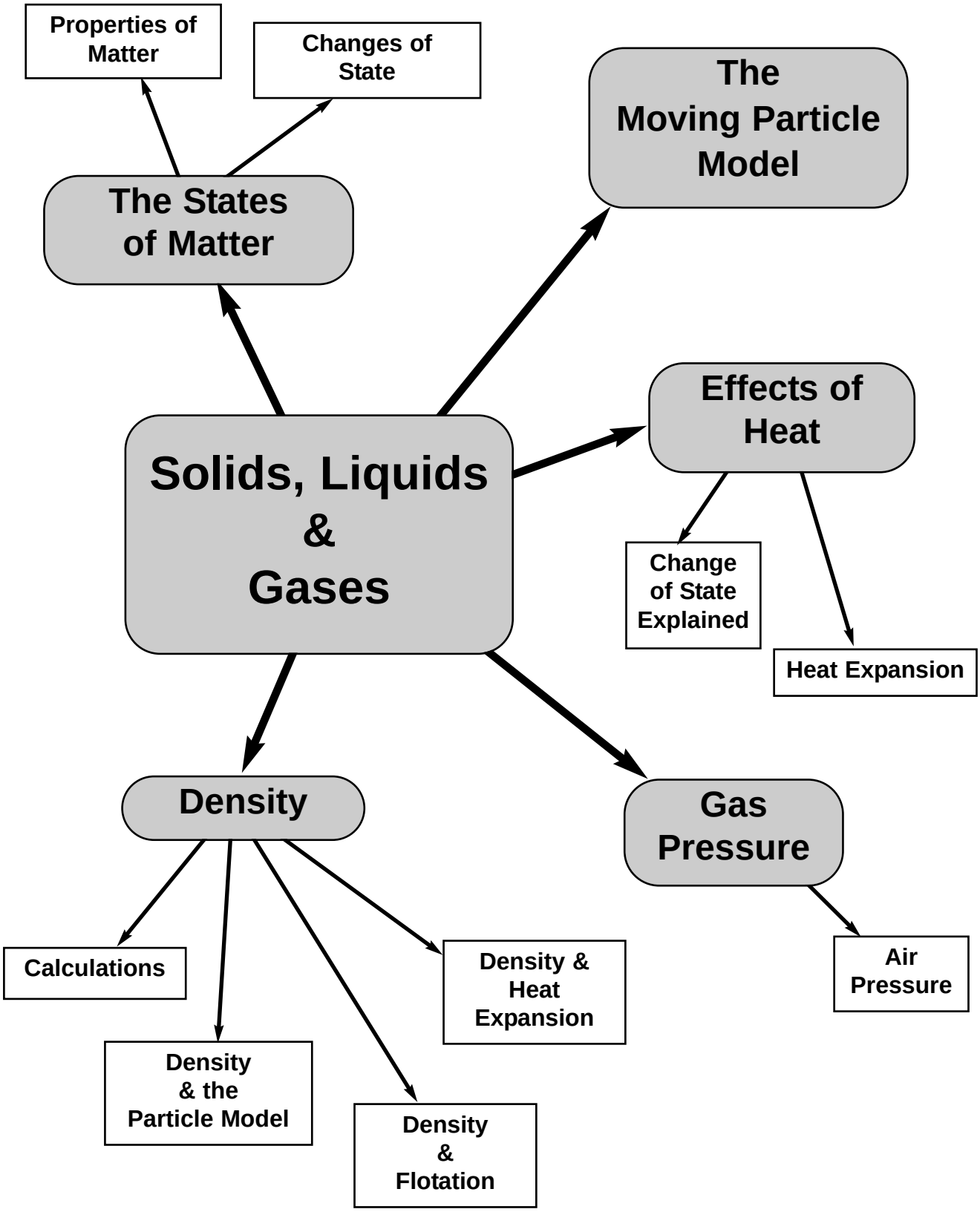
All Topics Available as PHOTOCOPY MASTERS and/or KCiC

Photocopy Masters (PDF files)
Black & White, A4 portrait-orientation for clear, economical photocopying.

KCiC = Key Concepts in Colour
Full colour, formatted for on-screen study and data projection. PDF + Powerpoint®
Powerpoint is a trademark of Microsoft Corp.

"Mind-Map" Outline of Topic

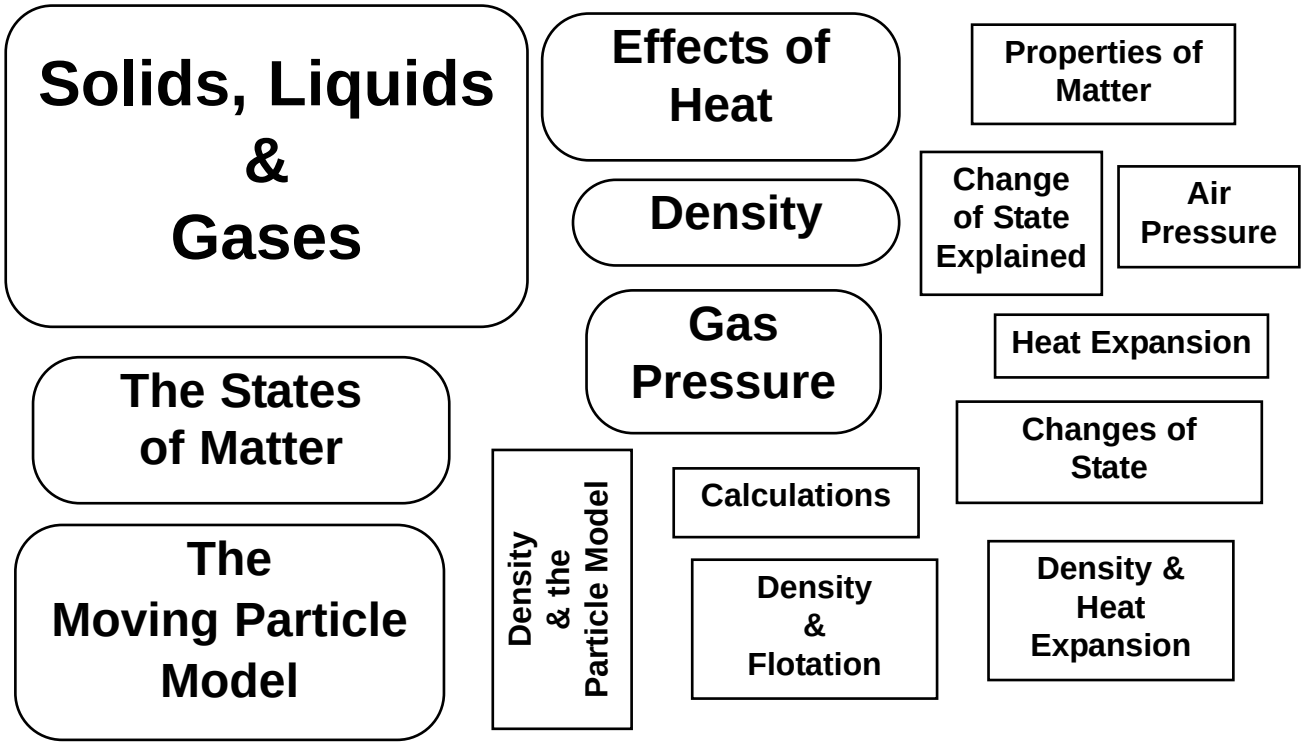
This topic belongs to the branch of Science called "Chemistry". Chemistry is the study of matter and materials. Chemistry looks at the properties of substances, and how substances can change into new forms.



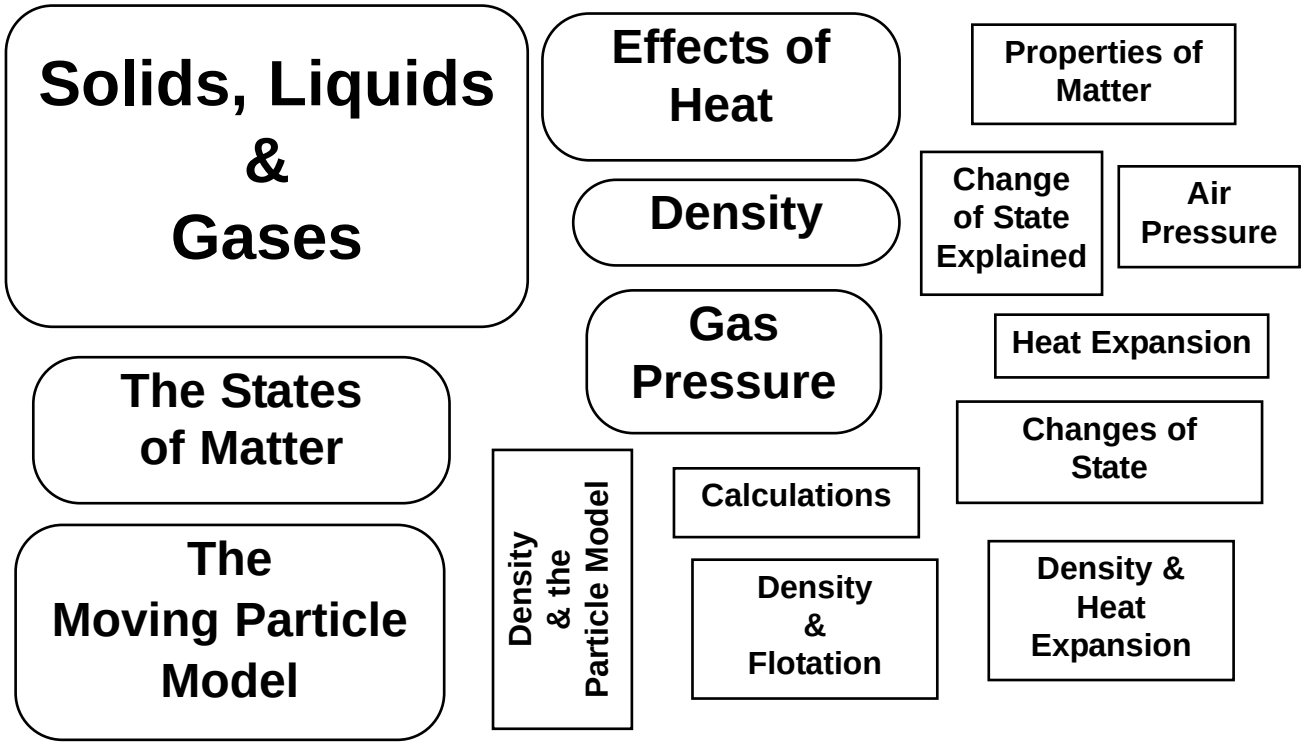
Usage & copying is permitted according to the Site Licence Conditions only

keep it simple science

Make your own "Mind-Map" TITLE PAGE.
Cut out the boxes. Sort them into an appropriate lay-out on a page of your workbook, then glue them down. Add connecting arrows and colour in.



Make your own "Mind-Map" TITLE PAGE.
Cut out the boxes. Sort them into an appropriate lay-out on a page of your workbook, then glue them down. Add connecting arrows and colour in.



copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

Usage & copying is permitted according to the Site Licence Conditions only

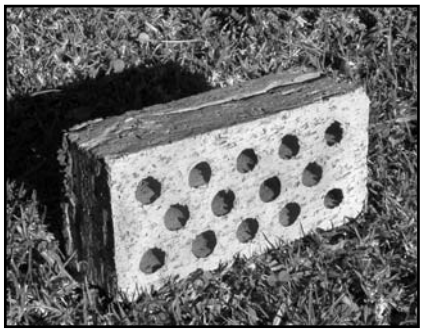
keep it simple science

Solid, Liquid, Gas

The Earth is mostly <u>solid</u> rock. The oceans are <u>liquid</u> water. The air we breathe is a <u>gas</u>.

All around us are many different substances, but when we come to study the science of matter, one simple starting point is that all substances are either solid, liquid or gas.

The States of Matter

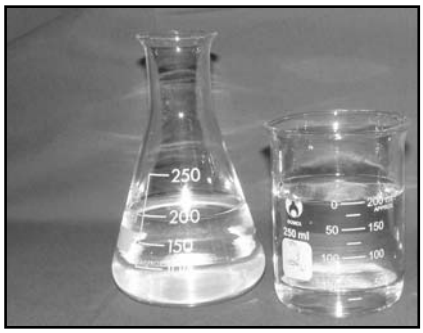


Solids

Generally,

- **have a definite shape.**
- **are hard.**
- **cannot be compressed.**
  (squashed into a smaller space)

A solid made of grains, like dry sand, has no definite shape and can flow like a liquid. However, each grain is hard, has a definite shape and cannot be compressed.



Liquids

Generally,

- **have no definite shape.**
  They take the shape of whatever container they are in.
- **can flow, and be poured.**
- **cannot be compressed.**



Gases

Generally,

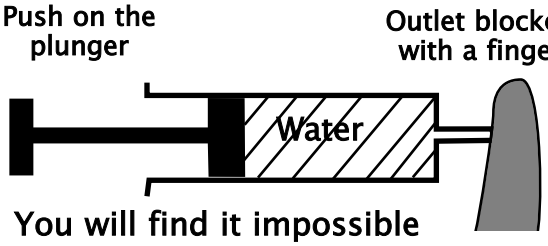
- **have no definite shape.**
  They completely fill whatever container they are in.
- **can easily flow and move.**
  (e.g. when the wind blows)
- **are easily compressed.**

Liquids & Gases are both "fluids"; substances which can flow and change shape.

Testing Compressibility

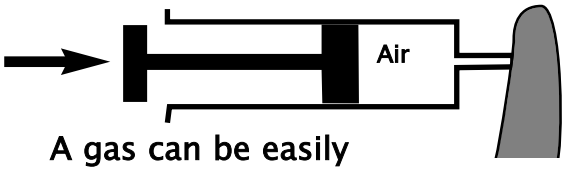
One of the big differences between the states of matter is whether or not a substance can be squashed into a smaller space.

This property can be studied using a syringe, or even a bike pump.



You will find it impossible to compress water

If you try again with the syringe filled with air, you'll find it very easy to compress the gas.



A gas can be easily compressed

<u>Simplest definitions of the States of Matter</u>

Solid = incompressible non-fluid
Liquid = incompressible fluid
Gas = compressible fluid

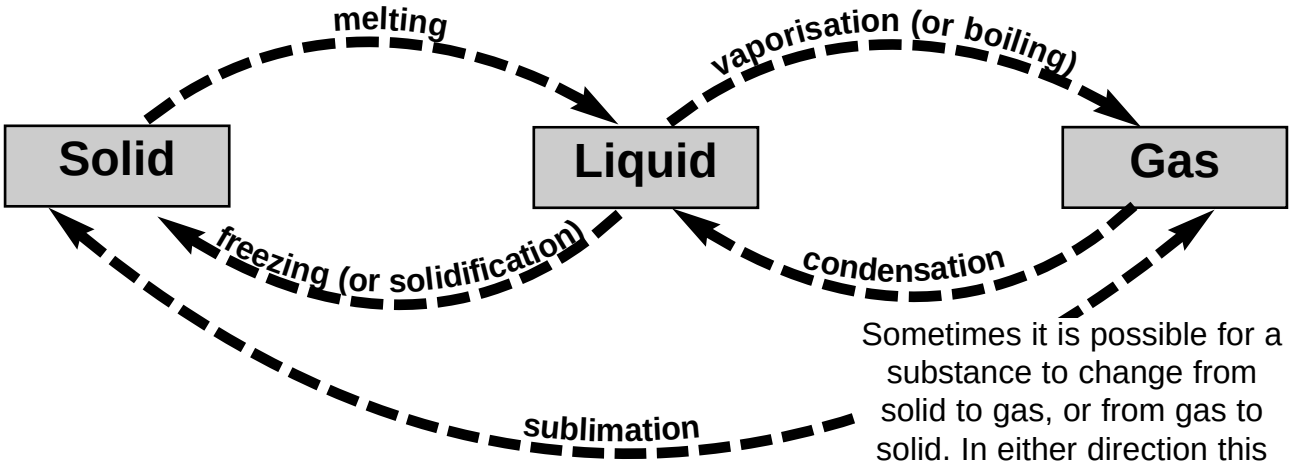
copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

Usage & copying is permitted according to the Site Licence Conditions only

keep it simple science

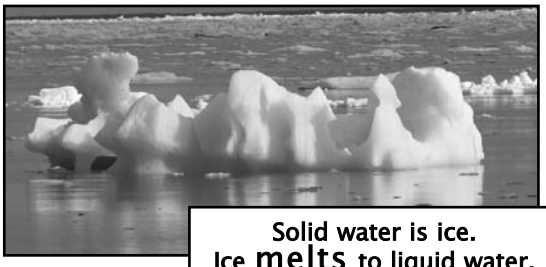
Changes of State

Most pure substances can change their state from solid to liquid to gas, and back again. When this happens, it is still the same substance, but in a different state.
LEARN THE NAMES FOR EACH CHANGE.

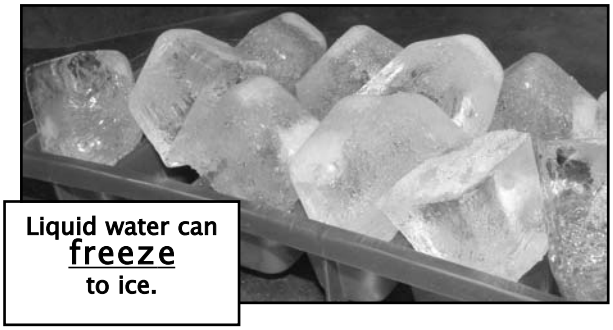


Sometimes it is possible for a substance to change from solid to gas, or from gas to solid. In either direction this change is called sublimation

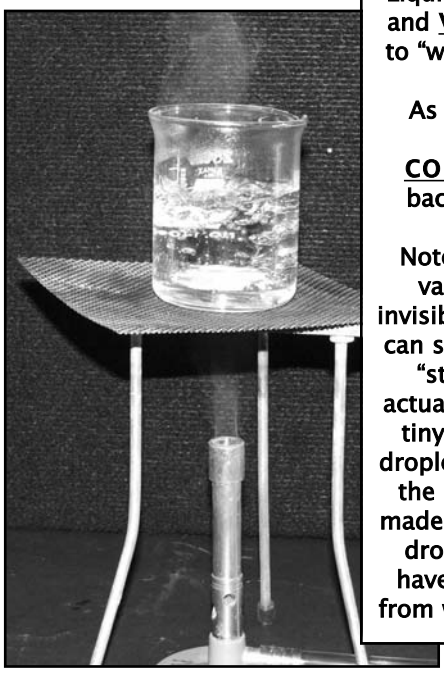
Examples of Changes of State



Solid water is ice.
Ice melts to liquid water.



Liquid water can freeze to ice.



Liquid water boils and vaporises to "water vapour".

As the vapour cools, it condenses back to liquid.

Note that water vapour is an invisible gas. If you can see a cloud of "steam", it is actually a cloud of tiny little liquid droplets. Clouds in the sky are also made of tiny liquid droplets which have condensed from water vapour.

Ice is solid water, but it is still the substance water.

Water vapour is a gas, but it is still water.

When a substance changes its state, it remains the same substance. It can change its state over and over again, but it is still the same stuff.

copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

Usage & copying is permitted according to the Site Licence Conditions only

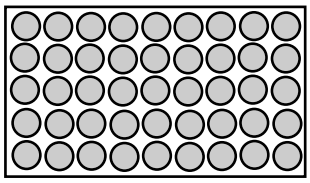
keep it simple science

The Moving-Particle Model

How can the properties of solids, liquids and gases be explained?

We believe that all substances are made of tiny lumps of matter we call <u>particles</u>. Each particle can be imagined to be like a tiny solid ball, too small to see.

Solid



The particles are tightly packed together.

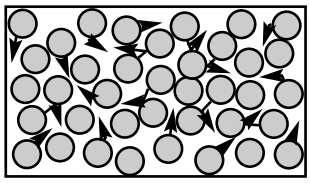
This explains why solids are incompressible.

They are held firmly in place by forces of attraction.

This is why solids are hard with a fixed shape.

Although the particles are fixed in place, they vibrate and jiggle around a bit.

Liquid



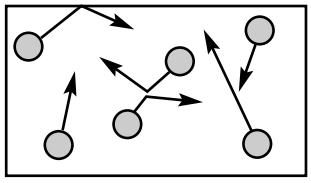
The particles are tightly packed together.

This explains why liquids are incompressible.

The particles are able to move around, bumping and jostling each other.

This explains why liquids have no fixed shape, and take the shape of their container.
This also explains how they can flow as a "fluid".

Gas



The particles are far apart.

This explains why gases are easily compressible. It is easy to push the particles closer together.

The particles are flying in all directions at high speed, colliding and bouncing away again.

This explains why they have no fixed shape, and totally fill their container.

Notice that in every case the particles are moving.
In solids, they only vibrate in one place. In liquids they move around among each other. In gases they fly freely at high speed.

Scientific Models

Is this description of solids, liquids and gases real? Well, yes, and no.

We have many good reasons to believe that all matter is made of small particles called <u>atoms</u>. Often they join together in larger lumps called "<u>molecules</u>".

However, we also know that atoms are not solid ball-like particles, but are made of many smaller pieces.

There <u>are</u> forces of attraction between atoms and molecules.

So overall, the description above is partly true, but is not the full story.

Scientists often develop "<u>models</u>" of things that cannot easily be seen. This helps in understanding and explaining the observed facts.

So long as the model explains things, and always remains totally in agreement with what we can see and measure, then it is useful, even if it is a simplification or not quite the full story.

For now, ball-like "particles" will be our model of matter.

copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

Usage & copying is permitted according to the Site Licence Conditions only

keep it simple science

Worksheet 1

States of Matter, Particle Theory

Fill in the blank spaces

Student Name.....

The 3 "states of matter" are a)....., ... and .....

Generally, solids are hard, with a definite b)..... and c)..... (can/cannot) be compressed.

Liquids have no definite shape, but take the shape of their d)..... Liquids can e)..... and be poured. Like solids, they cannot be f).....

Gases have no shape, and completely g)..... their container. They can flow and are easily h).....

When a solid turns to a liquid the change is called i)..... The reverse change (liquid to solid) is called j).....

When a liquid turns to a gas the change is called k).....

The reverse change (gas to liquid) is called l).....

It is also possible for a solid to turn to gas directly, without going through the m)..... stage. This change is called n).....

The "Moving o)..... Model" states that all substances are made of tiny particles.

In a solid, the particles are p)..... and held firmly in q)..... The particles cannot move around, but can r)..... back-and-forth.

In a liquid, the particles are also very s)....., but can t).....

In a gas, the particles are u)..... and are v)..... at high speed.

Worksheet 2

States of Matter, Particle Theory

Briefly answer each question

Student Name.....

1. Which feature of a gas:
a) allows it to be easily compressed?

b) causes it to always fill its container?

2. Which <u>two</u> states of matter are considered to be "fluids"?

3. Which feature of a solid:
a) makes it incompressible?

b) causes it to have a fixed shape?

4. Which feature of a liquid:
a) allows it to change its shape and flow?

b) makes it incompressible?

5. Which <u>two</u> states of matter cannot be easily compressed?

6. Name the change of state.
a) gas to liquid .....

b) liquid to solid .....

c) liquid to gas .....

copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

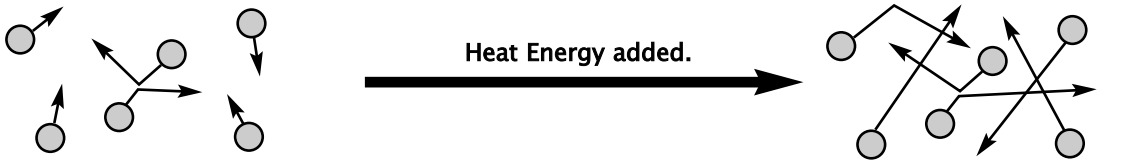
Usage & copying is permitted according to the Site Licence Conditions only

keep it simple science

Moving Particles? What Makes Them Move?

The answer is "<u>Heat Energy</u>".

When you add heat energy to any substance, the particles in it do not get hotter... they get faster. When something cools down, the particles do not get any colder... they go slower.



Moving particles in a substance.

Particles move faster. Temperature has increased.

The measurement we call "temperature" is actually a measure of how fast (on average) the particles are moving.

In a solid substance the particles cannot move around, but only vibrate. When a solid is heated, the particles vibrate faster, but still stay in their fixed place.

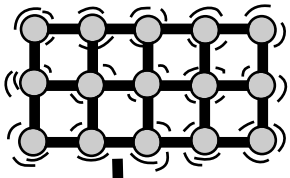
<u>Note</u>: The energy of a moving thing is called "<u>Kinetic Energy</u>". The Moving-Particle Model is sometimes called the "Kinetic Theory (Model) of Matter".

Changes of State... Again

With a knowledge of what heat energy does to particles, we can explain what happens in a change of state.

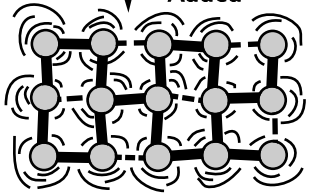
Melting

Particles in a solid are vibrating, but are held in place by forces of attraction.



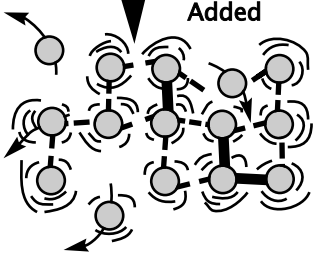
Heat Added

Adding heat makes the particles vibrate faster. The forces between them are still there, but the particles have more energy so the forces are almost overcome.



More Heat Added

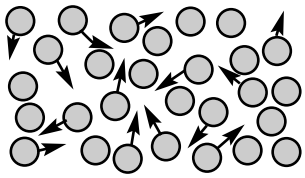
At a certain temperature (the "melting point") the particles break free from their positions and begin moving around. The forces are still there, but unable to hold them. The particles are still close together, but moving among each other.



The solid has melted!

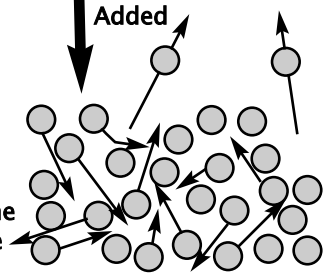
Evaporation

Particles in a liquid are close together, but move around among each other.



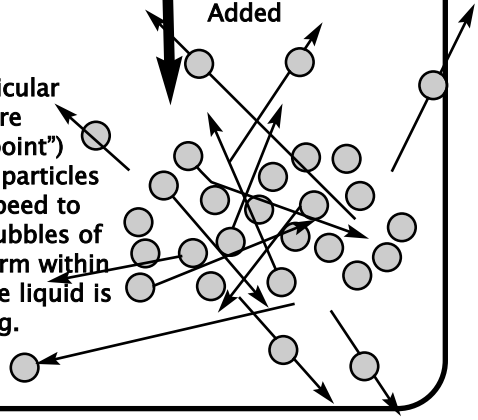
Heat Added

When heated, the particles move faster. Even at relatively low temperatures, a few particles have enough energy to fly off into the gas state. Some of the liquid is evaporating.



More Heat Added

At a particular temperature (the "boiling point") many of the particles reach the speed to evaporate. Bubbles of gas vapour form within the liquid... the liquid is boiling.



copyright © 2008 keep it simple science

Usage & copying is permitted according to the Site Licence Conditions only

keep it simple science

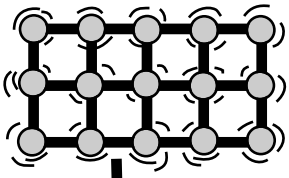
Heat Expansion

It is an observed fact that substances expand (get bigger) as they get hotter, and contract (shrink) as they get cooler.

How does our Moving-Particle Model explain this?

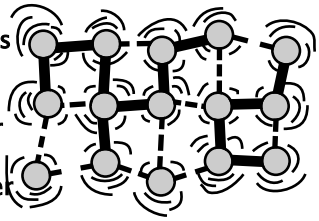
In a Solid...

Particles in a solid are vibrating, but are held in place by forces of attraction.



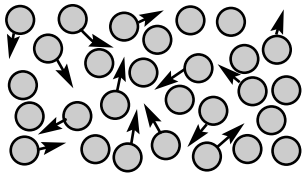
Heat Added

Adding heat makes the particles vibrate faster. As they jiggle and push against each other, they are forced a little further apart. This makes the solid substance get bigger and occupy more space.



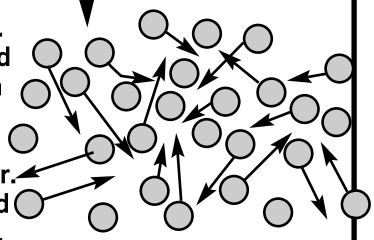
In a Liquid...

Particles in a liquid are close together, but move around among each other.



Heat Added

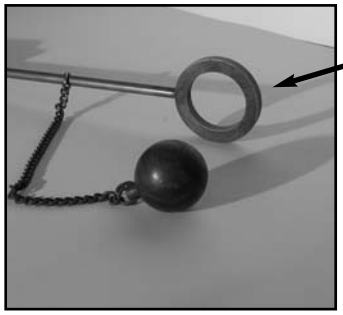
When heated, the particles move faster. They collide faster and harder and push each other away. This causes the volume of the liquid to get bigger. The liquid expands and occupies more space.



Very Important: Notice that the particles DO NOT expand. The particles force each other a bit further apart so the substance expands, but the particles stay the same size.

Investigating Heat Expansion

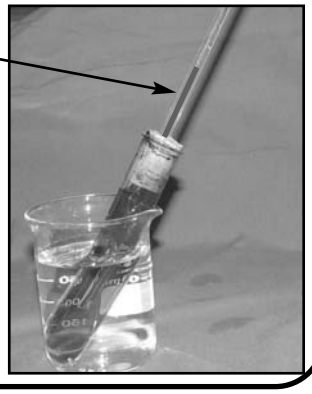
You may do some experiments, or see demonstrations, of heat expansion.



When both are cold, the ball fits through the ring.

When the ball is heated, it expands and won't fit through.

When warmed up, the liquid expands, so it rises up in the narrow glass tube.



This is the same principle as a thermometer

Heat Expansion in Structures

Heat expansion is a factor that must be allowed for when designing and building roads, railways, bridges and buildings.

On a hot day metal and concrete will expand, and will shrink again during a cold night. If this was not allowed for, the structure could warp or crack, and be damaged.

These structures are built with gaps, or flexible joints, to allow heat expansion to occur without damage.



The Sydney Harbour Bridge has a huge hinge at the base of the steel arch so the whole structure can flex and move.

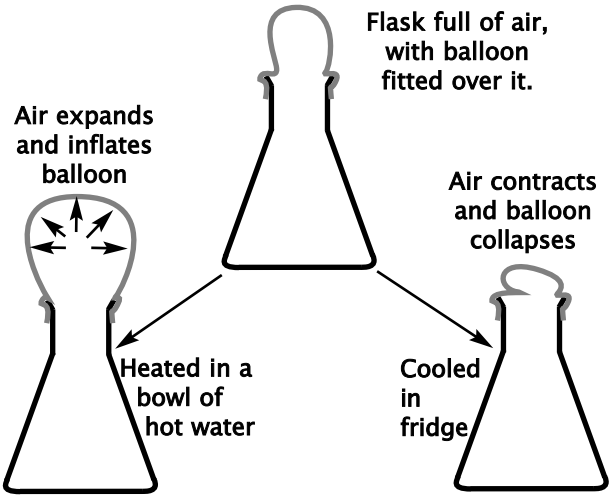
copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

Usage & copying is permitted according to the Site Licence Conditions only

keep it simple science

Heat Expansion in a Gas

You might also investigate heat expansion in a gas, as suggested by these diagrams.



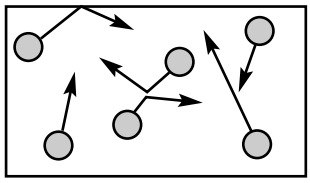
Expansion of a gas can be explained the same as before... particles move faster and push each other further apart.

However, what if the gas is inside a strong container that cannot expand the way a balloon can?

Now you must know about Pressure.

Gas Pressure

Remember that in a gas the particles are constantly flying everywhere at high speed.



They frequently collide with each other, and with the walls of their container. Each collision pushes on the inside of the container, or on each other. This means there is a constant push, or force, acting within the gas.

Pressure is measured as the amount of force pushing per unit of area.

The unit of pressure used most often in science is the **kilopascal (kPa)**

The "pascal" unit is named in honour of the great French scientist and mathematician Blaise Pascal (1623–62).

1 kPa = 1,000 N force per square metre.

Some examples of pressure values

Normal air pressure (sea level) ≅ 100 kPa

Inflated car tyre ≅ 200 kPa

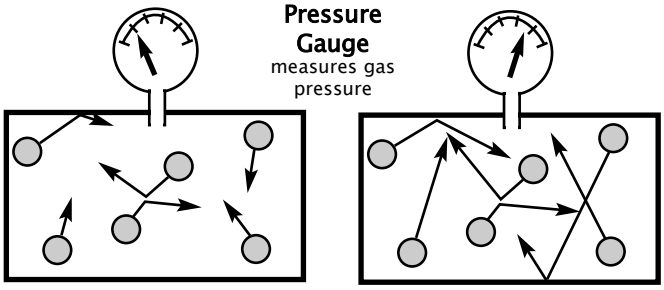
Inside a scuba airtank ≅ 800kPa

Changes in Gas Pressure due to Heating

What might happen if you heated a gas inside a strong container which cannot change its shape?

When heated, the particles move faster and push harder against each other and the container... so the pressure goes up.

OK, so it's a strong container, but there is a limit to how strong it is. If the pressure rises too high, the pressure can cause it to burst.



Gas in a rigid container, before heating.

After heating, particles move faster, pressure rises

This why you should never dispose of an aeorsol spray can in a fireplace or incinerator. As it gets hot the pressure will rise and it may explode.

Even more dangerous are pressure cylinders of fuel gas. If there is a fire in a house or factory, not only might the cylinder explode from rising pressure, but the escaping fuel will then burn. This is why large gas cylinders are always stored outside buildings.

copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

Usage & copying is permitted according to the Site Licence Conditions only

keep it simple science

Worksheet 3

Particles & Heat. Expansion & Pressure

Student Name.....

Fill in the blank spaces

When heat is added to any substance, the particles move a)..... When the substance is cooled, the particles move b).....

"Temperature" is really a measure of the average c)..... of the particles.

The melting of a solid occurs when heat causes the particles to d)..... fast enough to e)..... the forces holding them in place. They then begin to move around, so the substance is now a f).....

In a liquid, adding heat causes the particles to move around g)..... Some may have enough energy to fly off as particles of h).....

At the "boiling point", so many particles form vapour that i)..... of gas form within the liquid. We say that the liquid is "j)....."

Heat also causes most substances to k)..... because the faster moving particles push each other apart when they l)..... with each other.

If a gas is inside a container that cannot expand, then heating will cause the m)..... to rise. Pressure is caused by particles exerting a n)..... when they collide. Pressure is the amount of o)..... pushing per unit of p)..... The unit of pressure often used in Science is the q).....

Worksheet 4

Particles & Heat. Expansion & Pressure

Student Name.....

Match the Lists

Write the letter (A,B,C, etc) of the list item which matches each description. Not all the list items will be used, and <u>some may be used more than once</u>.

<u>Description</u>	matches with	<u>List Item</u>
1. When a substance gets bigger due to being heated.		.....
2. What happens to the actual particles when they are heated.		.....
3. Why liquid rises in a thermometer.		.....
4. What is inside the bubbles when a liquid is boiling.		.....
5. Why leave a gap in a rail track?		.....
6. Pressure is "force per unit of.."?		.....
7. Temperature at which bubbles of vapour form inside a liquid.		.....
8. Measurement of the average speed of particles in a substance.		.....

<u>List Items</u>

A. area
B. volume
C. contraction
D. melting point
E. heat expansion
F. boiling point
G. temperature
H. move faster
I. vapour (gas)

copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

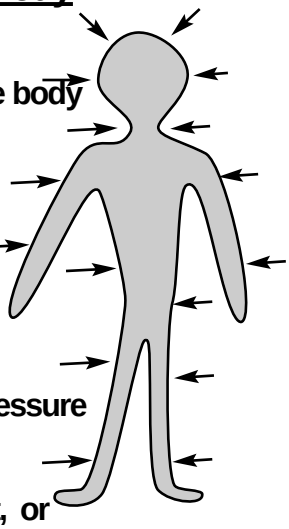
Usage & copying is permitted according to the Site Licence Conditions only

keep it simple science

Air Pressure

Air Pressure and Your Body

Normal air pressure at sea level is about 100 kPa. Over the area of your whole body this means a total force equivalent to about 1 tonne is pressing on you.



How come you don't feel squashed?

Simply because your body has the same amount of pressure inside, pushing outwards.

If you go up in an aircraft, or drive up a high mountain, the outside air pressure gets less. You will feel the pressure difference in your ears, until it equalises again.

Astronauts in space or on the Moon (no air) need pressure suits not just to breathe, but to protect them from the zero pressure around them. Their own "body pressure" pushing outwards would cause fatal damage without the suit.

Air Pressure and Weather

Most of our winds, storms and weather patterns are due to air pressure effects in the atmosphere.

In some places, a huge "bubble" of the air (maybe 1,000 km across) can warm up. This "bubble" expands so that the air particles become further apart and the air pressure becomes lower than the surrounding air. This is a "low pressure cell".

Meanwhile, in another place the air is cooling and contracting. The air particles are forced closer together, so the pressure rises... a "high pressure cell".

Both pressure systems begin to rotate because of the spin of the Earth, so circular winds blow around each "cell".

Low pressure cells often bring storms and rain (and cyclones) while "highs" tend to bring fine, dry weather. Both create the winds that blow around them, and from one to the other.

Tricks with Air Pressure

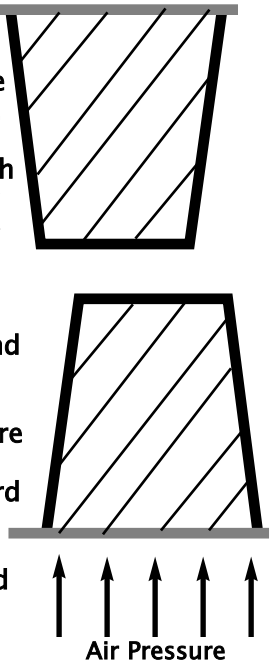
Totally fill a glass with water. Slide a piece of cardboard over it so there are no air bubbles inside.

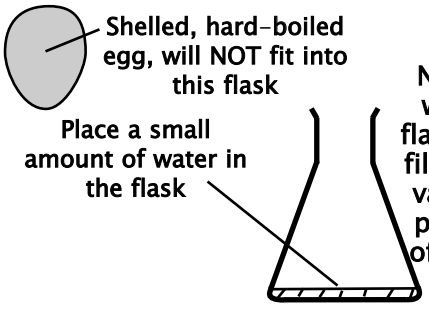
Support the cardboard with your hand while carefully turning the glass upside-down. Let go of the cardboard.

The water defies gravity and stays in the glass!

Explanation: the air pressure force on the cardboard is stronger than the downward weight of the water.

(But let air bubbles in, and see what happens!)





Turn off the heat, and immediately place the egg on the neck of the flask.

As the gas inside cools, and some water vapour condenses, the inside pressure drops.

Outside air pressure now pushes the egg into the flask.

Fascinating to watch!



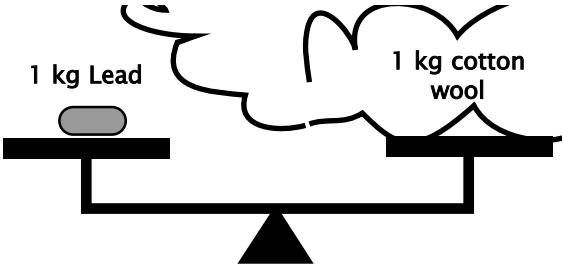
copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

Usage & copying is permitted according to the Site Licence Conditions only

keep it simple science

Density

Which is heavier?
1 kg of lead or 1kg of cotton wool?



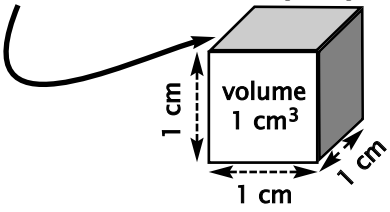
Well, of course, they are the same mass; 1 kg each. However, the lead has all its mass packed in a very small space or "volume", while the cotton wool occupies a large volume for the same mass.

The real difference here is "Density".

Density means the amount of mass per unit of volume.

In the laboratory, we would normally measure mass in grams (g).

Volume means the amount of space something occupies. Small solid objects would normally be measured in cubic centimetres (cm^3).





Liquid volumes are measured in millilitres (mL).

These 2 different units of volume are actually the same amount.

1 cm^3 = 1 mL

How to Calculate Density

Once the mass and volume of anything is measured, its density can be calculated as follows:

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

$$D = \frac{m}{V}$$

If mass is in grams, and volume in cm^3, then density will be in grams per cm^3 (g/cm^3)

Other possible units include g/mL and kg/m^3

Example Calculations

1.
A piece of lead has a volume of 2.5 cm^3 and a mass of 27.5 g.
What is its density?
Solution: D = m / V
= 27.5 / 2.5
= 11.0
∴ density is 11.0 g/cm^3

2.
A pack of cotton wool has a mass of 20g and it occupies a volume of 500 cm^3.
What is its density?
Solution: D = m / V
= 20 / 500
= 0.04
∴ density is 0.04 g/cm^3

copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

Usage & copying is permitted according to the Site Licence Conditions only

keep it simple science

Worksheet 5

Skills Exercise
Calculating Density
(calculator needed)

Student Name.....

For each question show full working. Write your answer rounded to 1 decimal place. Every answer must have units.

1.
A block of wood has a mass of 82g and a volume of 105 cm^3.
What is its density?

D = m / V

= ..... / .....

= .....

∴ density is ....
(answer) (units)

2.
A gold nugget (worth about $8,000) has a mass of 300g and a volume of 15.5 cm^3.
What is its density?

3.
A bottle of motor oil is labelled "500 mL". The oil in it has a mass of 400g.
What is its density?

4.
The density of a piece of steel = 8.0 g/cm^3. If the piece has a mass of 32g, what is its <u>volume</u>?

5.
Aluminium has a density of 2.7 cm^3. What <u>mass</u> is contained in a volume of 10 cm^3?

6.
Polystyrene foam plastic has a density of 0.2 g/cm^3. What <u>volume</u> of it do you need to have a mass of 1kg (1,000g)?

copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

Usage & copying is permitted according to the Site Licence Conditions only

keep it simple science

Worksheet 6

Skills Exercise (2 pages)
Calculating Volume & Density
(calculator needed)

Student Name.....

Part A

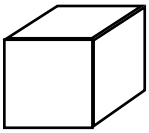
The volume of a rectangular prism or cube can be found by multiplying length x breadth x height. (V = L x B x H)

For each of the following

a) calculate the volume (in cm^3)
b) calculate the density

1.

This <u>cube</u> of copper is 2.0 cm on each side.



Its mass is 72g.
What is its volume, and its density?

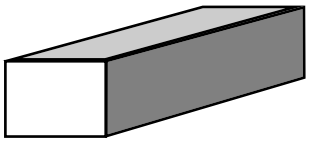
a) V = LxBxH **b) D = m / V**

= .....x.....x..... = .../ .....

= ..... = ...

Vol is .... density is .....
(answer, units) (answer,units)

2.

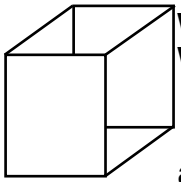


This brick measures 22.5cm x 11.0cm x 7.5cm.

Its mass is 5 kg (5,000g).
Find its volume and density.

3.

This empty box is 50cm high, 30cm wide and 75cm long.



What is its volume?

What would be the <u>mass</u> of a block of concrete exactly the same size as the box?

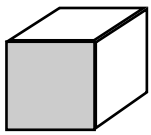
(density of concrete = 3.5 g/cm^3)

4.

A 750g box of breakfast cereal measures 21 x 13 x 9 cm.
Find its volume and density.

5.

This a <u>cube</u> of brass.



Its density is 6.5 g/cm^3 and its mass is 175.5g.
Find its volume. How long is each side?

copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

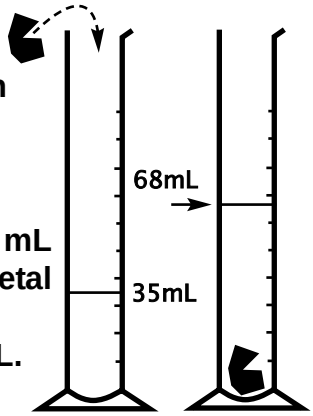
Usage & copying is permitted according to the Site Licence Conditions only

Worksheet 6 (cont)

Part B
Another way to measure volume is to lower a solid into water in a measuring cylinder. The water level rises.

The change in the cylinder reading is equal to the volume of the solid.

1.
A lump of metal with mass 198g was lowered into a measuring cylinder. The cylinder had 35 mL of water in it. The metal caused the water level to rise to 68 mL.



a) What is the volume of the metal?

b) What is its density?

2.
Another lump of the <u>same metal</u> (same density!) caused the cylinder reading to go from 42 mL up to 55 mL.
a) Volume of the metal?

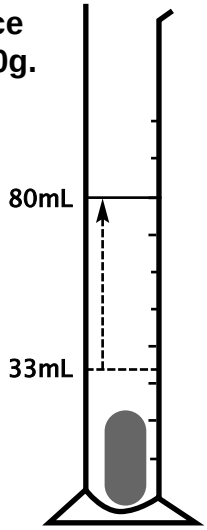
b) What is the <u>mass</u> of this lump?

3.
Yet another lump of this same metal was lowered into a cylinder containing 30mL of water. The metal had a mass of 252g.

a) What is the volume of this lump?

b) What would be the final reading of the measuring cylinder?

4.
Find the density of this piece of solid rock. Its mass = 120g.



Questions for Discussion
Some Limitations of this Method
a) Would this method of measuring volume work for an object which floats in water?

b) Would this method work for a solid like a sugar cube which will dissolve in the water?

copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

Usage & copying is permitted according to the Site Licence Conditions only

keep it simple science

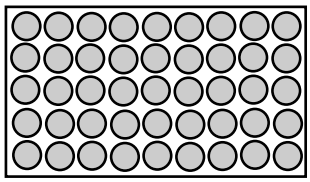
Density and the Moving-Particle Model

How does the Moving-Particle Model of matter explain density differences?

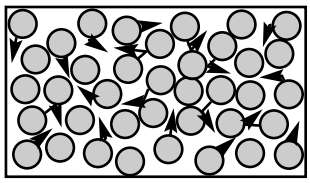
Solids, Liquids & Gases

Remember how the particles are arranged in the different states of matter.

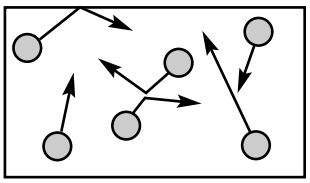
In solids the particles are packed close together, and fixed in place.



In liquids, the particles are also close together, but able to move around.



In gases, the particles are far apart and moving at high speed.



We could predict from the model that solids and liquids should have similar densities. Both solids and liquids should be far denser than a gas, assuming the particles are all the same weight.

Sure enough, if the density of any pure substance is measured the results are similar to the following.

Example	Density (g/cm^3)
Solid ice	0.92
Liquid water	1.0
Water vapour	0.0007

Note that the solid and liquid densities are similar, and both much higher than the gas.

Water is unusual because the solid is less dense than the liquid.

In most substances the solid is slightly more dense because the particles just move a bit further apart in the liquid, as shown in the diagrams above.

The slightly weird behaviour of ice and water will be explained later.

Comparing Different Solids

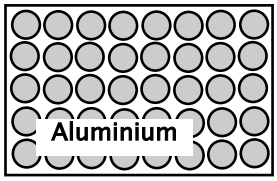
How does our model explain the different densities of different substances?

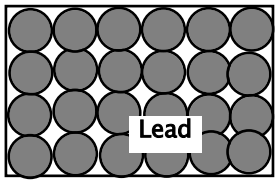
Particles Have Different Mass

Compare the metal aluminium with the metal lead:

	Density (g/cm^3)
Aluminium	3 (approx)
Lead	11 (approx)

Why such a difference?





Both types of particles pack together tightly. They are a bit different in size, but more importantly, each lead particle is much heavier than each aluminium particle. Lead is denser because its particles are heavier.

Particles Pack Differently

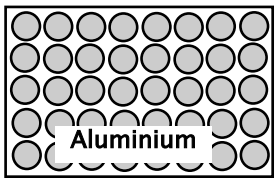
Compare aluminium with silicon:

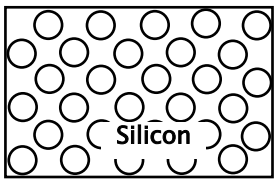
	Density (g/cm^3)
Aluminium	3 (approx)
Silicon	2 (approx)

In this case the particles of these 2 substances are almost exactly the same size and weight.

So why are their densities different?

It's because of the way their particles pack together.





The particles of silicon do not pack as tightly together as those of aluminium, so the density is lower.

The density of every substance depends on the mass of its particles, and how they pack together.

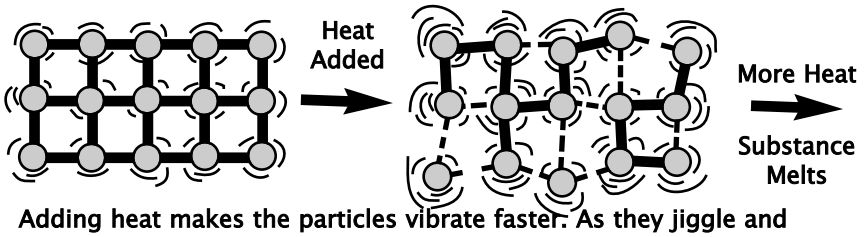
copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

Usage & copying is permitted according to the Site Licence Conditions only

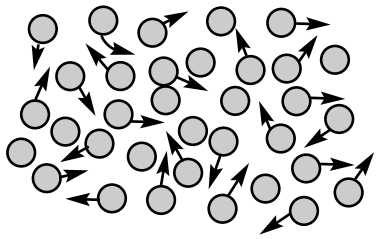
keep it simple science

Density and Heat Expansion

You've already learnt about heat expansion. What does that do to density?
Generally, as a substance gets hotter its particles move faster.
They collide harder and more often, and push each other apart slightly



Adding heat makes the particles vibrate faster. As they jiggle and push against each other, they are forced a little further apart. This makes the solid substance get bigger and occupy more space.



Particles in a liquid are still close together, but usually a bit further apart than in the solid.

When heat expansion happens, the substance (solid or liquid) gets bigger and has slightly more volume.
Since D = m/V, if the volume increases, but mass is still the same, then density must get less.

(However, these changes in density are only slight compared to the huge difference from liquid to gas.)

Water the Weirdo!

Most of the time, water acts just like all other substances.

For example here are its density values over a range of cooling temperatures.

Temperature (°C)	Density (g/cm³)
99	0.96
75	0.98
50	0.99
25	0.997
5	1.00

Like other substances, as you cool water down it shrinks slightly and its density rises a little.

However, when you cool other substances further until they freeze (solidify) they keep shrinking and density continues to rise. But look at what happens to water:

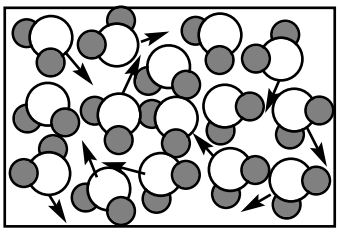
Temperature (°C)	Density (g/cm³)
5	1.00
0 (still liquid)	0.99
0 (ice)	0.92

As water freezes, it expands, and the density gets lower! Why?

Why Water Expands on Freezing

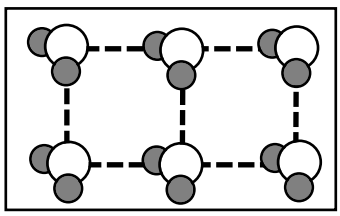
This is tricky to explain. You need to know that the "particles" in water are not actually ball-like, but shaped like this:





In liquid water, these particles can wriggle in very close to each other for maximum density.

However, as water freezes to ice the particles are forced to line up in organized "ranks", so they are actually slightly further apart.



This is why water expands as it freezes, and
(that's why a can of drink will split open if frozen)
ice floats in water (it has a lower density)

copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

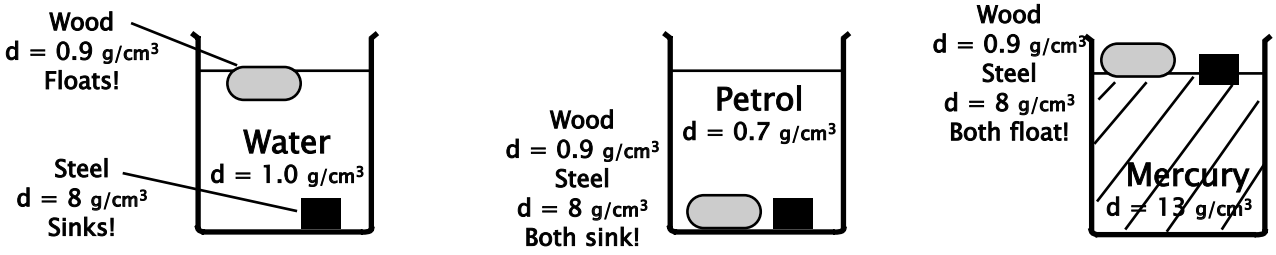
Usage & copying is permitted according to the Site Licence Conditions only

keep it simple science

Density and Flotation

Why do some things float and others sink?

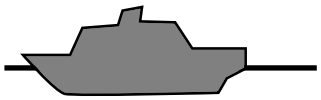
It's all a matter of density. Study these diagrams to spot the rule.



Can you see the pattern? Things will float if their density is less than the liquid. They will sink if their density is more than the liquid.

How Can a Steel Ship Float?

Because of its shape, the ship contains a lot of air within its total volume. This means its overall density is only about 0.7 g/cm^3.



Since fresh water has d = 1.0 g/cm^3, the ship floats.



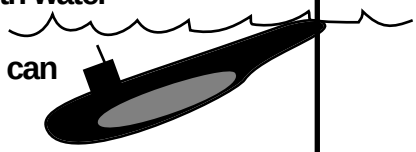
It floats even better in sea water which has a higher density. (salt water, d = 1.3 g/cm^3)

If the ship gets a hole in it and enough air spaces fill with water, the density rises and it may sink.

A submarine has special ballast tanks. When these tanks are filled with air, the sub's density is less than water, so it floats.



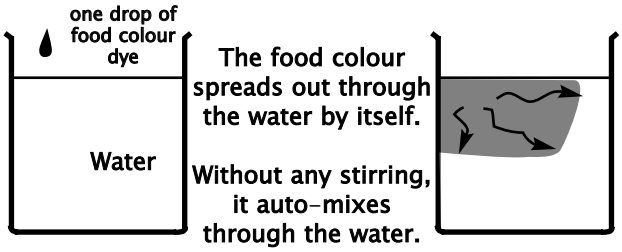
If the tanks are filled with water the sub's density is higher than water, so it can dive.



Cruising underwater, the tanks are adjusted so that the sub has the same density as the sea... "neutral buoyancy". In this state it neither sinks nor floats, but hangs in the water, and it can manoeuvre easily.

Diffusion

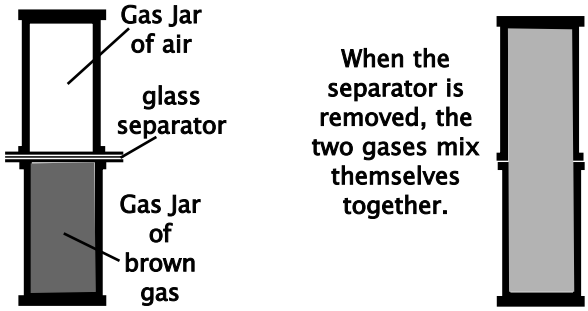
You might do one of these activities yourself, or see it demonstrated.



Fluids (liquids and gases) seem to be able to mix themselves together automatically.

This process is called "Diffusion".

The coloured solution can mix itself through the beaker of water. The brown gas can mix with the air without any help.



The explanation is in the Moving-Particle Model of matter. In liquids and gases, the particles are moving around. If 2 different gases or liquids are side-by-side, then the moving particles will automatically mix.

Is diffusion faster in liquid or gas?
What effect would temperature have?

copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

Usage & copying is permitted according to the Site Licence Conditions only

keep it simple science

Worksheet 7

Density & The Particle Model

Fill in the blank spaces

Student Name.....

In both a)..... and ..... the particles are very close together, compared to particles in a b)..... This means that the density of solids and liquids is much c)..... (higher/lower) than a gas.

For most pure substances the density of the solid is slightly d)..... (higher/lower) than the liquid because as the substance melts, the particles get a bit e)..... (further apart/closer together).

Different substances have different densities because their particles may have different f)..... , or because their particles may g)..... in a different way.

When heat expansion occurs in a solid or liquid, the substance gets bigger so it occupies a larger h)..... This means its density i)..... (increases/decreases)

Water is an unusual substance because it j)..... as it freezes. The result is that ice is k)..... (more/less) dense than water, so ice will l)..... in water.

Whether something floats or sinks depends on density. If an object is more dense than the liquid, it will m)..... If it is less dense than the liquid it will n).....

A ship is built from high-density steel floats because its overall density is o)..... than water due to its shape and the many p)..... spaces it contains. By adjusting the amount of water or air in its "q)..... tanks", a submarine can dive, surface, or "hang" in the water with "r)..... buoyancy".

Diffusion is when s)..... or ..... can mix themselves together. It happens because the particles are t)..... so they automatically mingle with each other.

Worksheet 8

Sink or Float?

Student Name.....

Here are the densities of 3 different liquids, and 4 different solids.
(all in g/cm^3)

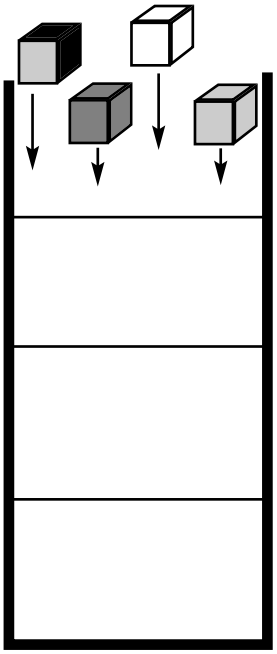
Liquids		Solids	
water	1.0	lead	11
mercury	13.5	plastic	0.75
olive oil	0.8	uranium	19
		wood	0.9

The 3 liquids were poured into a container together. They did NOT mix, but formed layers as shown in the diagram.

a) label the layers to show where each liquid ended up.

Next, small cubes of the 4 solids were dropped in.

b) Sketch on the diagram to show where each one ended up.
(label them)



copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

Usage & copying is permitted according to the Site Licence Conditions only

Topic Test
Solids, Liquids & Gases

Answer all questions
in the spaces provided

Student Name.....

Score = ... / 40

1. (10 marks)
Match each description to an item from the list. To answer, write the letter (A,B,C, etc) of the list item beside the description.

Description matches with **List Item**

i) The lowest density state of matter. .....

ii) Easily compressed fluid. .....

iii) Change of state from gas to liquid. .....

iv) Change of state from liquid to solid. .....

v) The state in which particles are held in one place. .....

vi) Energy which causes particles to move faster. .....

vii) Force per unit of area. .....

viii) Mass per unit of volume. .....

ix) Process of fluids mixing themselves together. .....

x) Incompressible fluid. .....

List Items Not all will be used. Some may be used more than once.

A. solid
B. liquid
C. gas
D. evaporation
E. diffusion
F. density
G. condensation
H. pressure
I. freezing
J. heat

2. (6 marks)
In the boxes below, use simple sketches to represent a model of the particles in solids, liquids and gases.

In each sketch you must make it clear:
- **if particles are close, or far apart.**
- **if particles are able to move around**

Solid

Liquid

Gas

3. (6 marks)
Briefly explain:
a) what happens to the particles in a solid when the substance melts to become liquid.

b) why a substance expands when heated.

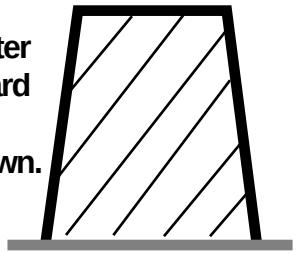
c) why the gas pressure increases in a sealed container when it is heated.

copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

Usage & copying is permitted according to the Site Licence Conditions only

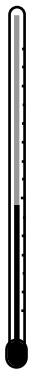
4. (4 marks)

a) This is a glass of water with a piece of cardboard on top, which has been turned upside-down.



Explain why the water does not fall out of the glass.

b) When the temperature increases, the liquid in this thermomter rises up inside the glass tube.



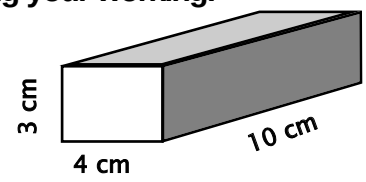
Explain why the liquid rises.

5. (6 marks)

For each of the following, calculate the density of the object shown or described.

Marks will be awarded for showing your working.

a) mass = 240 g



b) A lump of metal with mass 150g was dropped into a measuring cylinder of water. The water level rose from 40 mL to 55 mL.

6. (3 marks)

Carbon exists in several different forms. Diamond has a density = 3.5 g/cm^3.
Graphite has a density = 2.3 g/cm^3.
Both forms are made up of exactly the same particles (same atoms).

a) Suggest a reason why they might have different densities.

b) If you heated a diamond up, what would you expect to happen to its density as it got hotter?

7. (5 marks)

Fill in the blank spaces.

An object will float in a liquid if the object's density is a)..... the density of the liquid.

If you cool a liquid until it turns solid, usually its density will b)..... as it solidifies. Water is unusual because ice is c)..... dense than liquid water. This is because, as it freezes, the particles in water re-arrange so they are d)....., and the volume of the water e)..... as it freezes.

copyright © 2008 keep it simple science
www.keepitsimplescience.com.au

Usage & copying is permitted according to the Site Licence Conditions only

Answer Section

Worksheet 1

a) solid, liquid & gas
b) shape
c) cannot
d) container
e) flow
f) compressed
g) fill
h) compressed
i) melting
j) freezing
k) evaporation
l) condensation
m) liquid
n) sublimation
o) Particle
p) close together
q) place
r) vibrate
s) close together
t) move around
u) far apart
v) moving

Worksheet 2

1.
a) Particles are far apart, with a lot of empty space between them.
b) Particles are flying at high speed in every direction.

2.
Liquids and gases

3.
a) Particles are very close together and cannot easily be pushed any closer.
b) Particles are fixed in place and held by forces of attraction.

4.
a) Particles are not fixed in place and can move around.
b) Particles are very close together and cannot easily be pushed any closer.

5.
Solids and liquids

6.
a) Condensation
b) Freezing
c) Evaporation

Worksheet 3

a) faster
b) slower
c) speed
d) vibrate
e) overcome (break free of)
f) liquid
g) faster
h) gas
i) bubbles
j) boiling
k) expand
l) collide
m) pressure
n) force
o) force
p) area
q) kilopascal (kPa)

Worksheet 4

1. E
2. H
3. E
4. I
5. E
6. A
7. F
8. G

Worksheet 5

1.
D = m / V
= 82 / 105
= 0.7809...
density is 0.8 g/cm^3.

2.
D = m / V
= 300 / 15.5
= 19.354...
density is 19.4 g/cm^3.

3.
D = m / V
= 400 / 500
= 0.8
density is 0.8 g/mL.

4.
D = m / V
8.0 = 32 / V
so V = 32 / 8
= 4.0
volume is 4.0 cm^3.

5.
D = m / V
2.7 = m / 10
so m = 2.7 x 10
= 27.0
mass is 37.0 g.

6.
D = m / V
0.2 = 1,000 / V
so V = 1,000 / 0.2
= 5,000
volume is 5,000 cm^3.

Usage & copying is permitted according to the Site Licence Conditions only

Worksheet 6

Part A

1.

a) V=LxBxH
=2x2x2
= 8
Vol is 8.0 cm^3.

b) D = m / V
=72 / 8
= 9
Density is 9.0 g/cm^3.

2.

a) V=LxBxH
=22.5x11.0x7.5
= 1856.25
Vol is 1,856.3 cm^3.

b) D = m / V
= 5,000/1856.25
= 2.693...
Density is 2.7 g/cm^3.

3.

a) V=LxBxH
=50x30x75
= 112,500
Vol is 112,500 cm^3.

b) D = m / V
3.5 = m / 112,500
so m =3.5x112,500
= 393,750
Mass is 393,750 g
(about 394 kg)

4.

a) V=LxBxH
=21x13x9
= 2457
Vol is 2,457 cm^3.

b) D = m / V
=750 / 2457
= 0.305...
Density is 0.3 g/cm^3.

5.

D = m / V
6.5 = 175.5 / V
So, V = 175.5 / 6.5
= 27
Volume is 27 cm^3
Since it is a cube, each side must be 3cm (3x3x3 = 27)

Part B

1.

a) V = 68-35 = 33 mL
b) D = m / V
= 198 / 33
= 6
density is 6.0 g/mL.

2.

a) V = 55-42 = 13 mL
b) D = m / V
6 = m / 13
So, m= 6x13
= 78
Mass is 78.0 g

3.

a) D = m / V
6 = 252 / V
So, V= 252/6
= 42
Volume is 42 mL.
b) Reading would rise to 72 mL. (30 +42)

4.

a) V = 80-33 = 47 mL
b) D = m / V
= 120 / 47
= 2.553...
density is 2.6 g/mL.

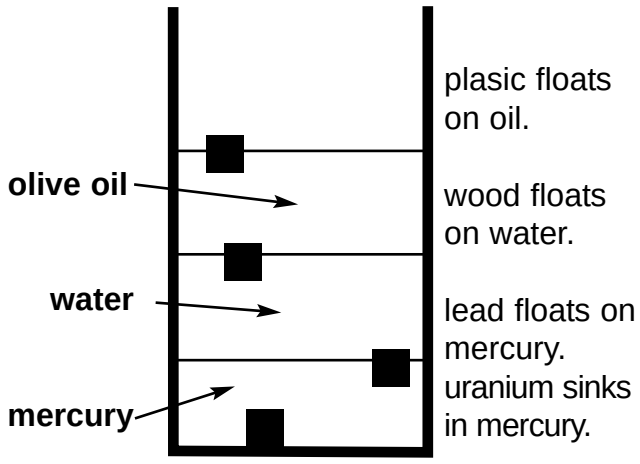
Questions

a) No, because if it is floating it hasn't "pushed aside" its own full volume of water. It must be completely under to do so.
b) No, it will not "push aside" the water. Its particles will mingle in with the water and not displace it.

Worksheet 7

a) solids & liquids **b) gas**
c) higher **d) higher**
e) further apart **f) masses (weight)**
g) pack together **h) volume**
i) decreases **j) expands**
k) less **l) float**
m) sink **n) float**
o) less **p) air**
q) ballast **r) neutral**
s) liquids or gases **t) moving around**

Worksheet 8



copyright © 2008 keep it simple science
www.keepitsimplescience.com.au
Usage & copying is permitted according to the Site Licence Conditions only

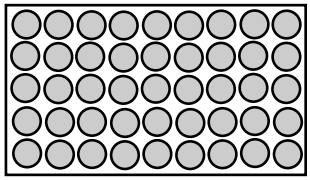
Topic Test

1.

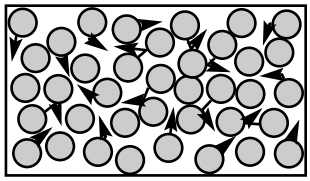
i) C ii) C iii) G iv) I v) A
vi) J vii) H viii) F ix) E x) B

2.

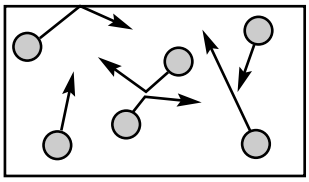
Solids
close together
fixed in place



Liquids
close together
moving around



Gas
far apart
flying everywhere



3.

a) Particles begin vibrating more and more until the forces cannot hold them in place. They start moving around.

b) Particles begin moving faster. They collide harder and push each other a bit further apart, so the substance gets bigger.

c) Particles move faster and collide more often and with more force. This force (per unit of area) is the pressure, so it goes up.

4.

a) the weight of the water is help up by the air pressure pushing on the cardboard.

b) Liquid rises because of <u>heat expansion</u> as the liquid gets hotter.

5.

V = LxBxH
= 10x4x3
V = 120 cm^3.

D = m / V
= 240 /120
= 2
Density is 2.0 g/cm^3.

b) Vol = 55-40 = 15 mL.
D = m / V
= 150 / 15
=10
Density is 10.0 g/mL.

6.

a) The particles (atoms) pack together differently. (In diamond they pack closer together)

b) Density will get lower. This is because heat expansion will make the volume increase, so density must decrease.

7.

a) less than
b) increase
c) less
d) further apart
e) increases

Usage & copying is permitted according to the Site Licence Conditions only